Neuroergonomics in Digital Operating Rooms: Applying the Two-Competitor Model of Attention to the Surgical Context

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***Abstract*—Due to high workload and often excessive working hours, team members in operating rooms perform surgical procedures under difficult conditions and often without sufficient breaks. Despite this, the team must deal with incomplete information and unexpected distractions. This requires a suitable level of attention and the ability to balance the demands of the task with available cognitive resources. Advances in measurement technology and data analysis in neurotechnology open up new possibilities for the assessment of attention processes. Increasingly complex and demanding surgeries, especially, could benefit from the application of neuroergonomic automated assistants to minimize distraction, stress, and fatigue, and to facilitate interactions between team members. Such assistants could improve performance via monitoring of cognitive and affective states as well as the implementation of suitable interventions strategies. Understanding the impact of distractions on performance, enhancing individuals' resilience to distractions, and potentially employing artificial assistants to mitigate their effects are critical future goals. In order to support such future developments, a standard taxonomy of attention in the operating theater is needed, as is a broader consensus regarding the nature of distraction. Ideally, such a model would serve as a basis for comparison between studies conducted in different laboratories, and in principle could also be used to bridge the gap between the laboratory and the real scenario. Here, we propose the adoption of a model of attention previously shown to be effective for modeling levels of attention in immersion and describe its application in the surgical context.**

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I. Introduction

DISTRACTION in surgery is a well-known problem and managing it is a key professional skill, associated with surgical inefficiency, cognitive overload, and stress [1], [2], [3], [4], [5], [6]. An attentive, undistracted team and in particular an undistracted surgeon is essential for an effective procedure, low human error rate, and eventually a positive outcome of a surgery [7]. It is important to be able to identify and correct factors that lead to distraction and impair the performance of surgeons both in training and under regular performance conditions. Monitoring attentional engagement and distraction during training in the classroom and the laboratory allows instructor and student to identify and remediate inefficient behaviors before they become ingrained. In the operating room (OR), with increasing technological development we can foresee the potential for using neuroergonomic cognitive assistants to support and facilitate attentional engagement during actual surgeries, especially under extenuating circumstances [8], [9], [10], [11], [12], [13], [14], [15]. However, for such efforts to be successful, we need validated and standardized tools for the assessment and mitigation of distractions in a surgical context, which are currently lacking [16].

Most available research on distraction in the OR is based on survey data from interviewing doctors or on analytic data from measurements obtained in simulated settings rather than real work environments [17], [18], [19], [20], [21], [22], [23], [24]. Thus far, very few studies involve actual measurements made in nonsimulated surgical settings with realistic demands on attention and distraction of surgeons [25]. This is not surprising given the complexity of the OR environment, involving the interaction of multiple team members, a wide variety of instruments and devices, and the unpredictability of the human body, combined with the difficulties of obtaining unobtrusive, objective measurements associated with attention and distraction even under optimal circumstances. Even in scenarios with realistic environments that are less complex, there is still a significant gap in the literature on unobtrusive affective sensing [26].

Research has made significant progress toward successful unobtrusive monitoring; however, there are certain limitations that

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hinder these subtle technologies from finding wider and more practical applications [27]. For instance, the sensing quality of nonintrusive physiological and psychophysiological monitoring systems might not be as good as medical-grade equipment. This could be attributed to wearable setups, which often sacrifice signal quality as they are primarily designed for extended usage and transmit data mainly through wireless channels. Despite these challenges, nonintrusive sensing has shown considerable promise in various settings. The benefits of these methods became increasingly evident during the COVID-19 pandemic [28], [29]. Still, the majority of technologies are unable to track physiological signals over long periods of time with sufficient reliability to be practical in an OR context. Because of activities or events that could inadvertently interrupt monitoring, remote monitoring technologies including remote photoplethysmography (rPPG), or video motion methods are currently mostly used for scientific validations against gold standard methods [26], [29]. However, it is important to consider how we might move beyond the status quo. To develop repeatable strategies for employing and evaluating monitoring methods, it is crucial to create a model of neuroergonomics in the OR, specifically mapping attention and distraction. Such a model could guide future research and evaluate the outcomes of various neuroergonomics studies in the surgical context using a unified theoretical framework. Here, we describe a model that has been previously developed to account for attention and distraction in complex, immersive environments [30] and consider how this model may be applied to guide future research on distraction and remedies for distraction in the OR. While this model is applied in a surgical context, its principles extend to other highly digitalized domains, such as Industry 4.0/5.0, where perceptual strain plays a role in human–human and human–machine teaming and distractions paired with fatigue and stress are critical factors to consider.

II. Distractions During Surgery

Survey-based research has revealed a variety of circumstances of distraction during surgery, and some supporting measurements describe the rate of the occurring distracting events and their impact of teamwork, stress, and workload [1], [2], [3], [6], [7], [16], [31], [32], [33], [34]. Notable distractors include OR door opening, device malfunctions, and irrelevant conversations, as well as more purely cognitive demands such as teaching trainees during surgery, mental preoccupation with management of the next case, and time pressure more generally. Surgeons have tight time schedules and often need to prioritize tasks, especially since many surgical tasks are mentally and/or physically very demanding. Many factors influence the mental/physical state of the surgeon and surgical team and can lead to unattended consequences and decrements in performance [35]. Attentional resource capacity, for example, is one of the key factors that limits performance even in very skilled surgeons [36], [37], as is fatigue [1], [17], [18], [38], [39], [40]. Both fatigue and the availability of attentional capacity may change over the course of a day or even a single surgery [41], meaning that complications resulting from immediate distraction in operating theaters can become very serious [16].

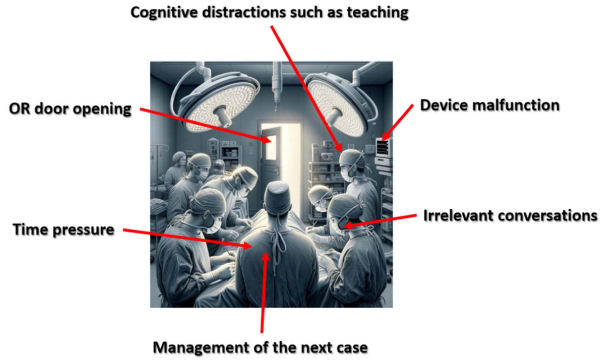


Fig. 1. Most distracting factors reported from OR team members in various surveys. | Created with DALL· E, an AI system by OpenAI

Only a few authors have described ways to mitigate the aforementioned problems [19], [35], [42]. There are potential technologies that might be used for attention assistance in operating theaters, including augmented reality glasses, bone-conduction headphones, or body interfaces for haptic information transfer [10], [15], [43], [44]. Some of them could even be used for sensing and sending important information from/to the individual team members. Furthermore, interbrain synchronization could provide useful information for comprehending team performance and problem solving as a whole [45], [46], [47], [48]. Without an in situ assessment of ongoing mental processes, it is impossible to evaluate individuals in group interaction patterns. Hyperscanning might be used to assess physical artifacts of mental activity continuously, which is an indirect way to achieve this goal [49]. However, both improving methods to study attention in situ and (eventually) being able to create effective external systems to mitigate the effect of distractors in the OR, require a precise understanding and description of the distractions and their effects on the OR team in action.

A. Sources of Distraction

To be able to implement interventions, it is important to identify the sources of distraction that shift the attention of surgeons and their team away from the primary task. Fig. 1 shows the factors that were reported to distract surgeons the most. These are findings from a human-factor survey conducted during 256 laparoscopic surgeries [50]. These results were confirmed by other studies investigating the event frequency during procedures and the perception of distractions among surgeons, anesthesia team members, nurses, and scrub technicians [31], [51]. Other surveys mentioned highest stress levels associated with distractions related to equipment failures and people entering or exiting the OR [32], [39].

Based on examination of previous literature together with observation reported to us by surgeons, we could group distractions in the OR according to a taxonomy that highlights the relationship between the source of the distraction and the individual (i.e., the surgeon) being distracted. Any model of attention during surgery must be able to accurately represent these kinds of distraction. These can be considered in terms of their origin, as follows:

1) from within individuals within the team (distractions, time pressure);
2) from outside the team (doors opening, device malfunction);
3) from within the team (irrelevant conversations).

1) Internal Distractors: With respect to internal distractions, these arise mainly as a result of specific factors affecting a given individual, especially uncertainty or time constraints. For example, lead surgeons are often also responsible for patients on the ward. This means that they know that their pager can alert them or the phone can ring at any time to get important information from them for situations occurring on the ward, especially if some of the patients there were already in difficult circumstances before the surgery started. Even in the absence of an actual pager alert (see *external distractions* below) the uncertainty associated with the possibility of an alert can play a role in the back of the surgeon's mind, causing an ongoing distraction (i.e., demand on cognitive resources). Furthermore, uncertainty can also arise from external circumstances outside the physician's control, such as whether crucial anatomical areas were sufficiently visible on the preoperative scans or if the patient is at higher risk for complications or other unexpected findings they could not prepare for. Similarly, time constraints and planning for the next case may contribute additional demands to the current procedure. All members of the team are likely to have factors that, while different in terms of the specific sources of uncertainty or time pressure, may still impose similar demands on cognitive resources (e.g., concerns about getting done in time to pick a child up from day-care, worries about the well-being of a previous patient, etc.). In the ideal case, each member should be able to cope with such internal sources of distraction, suppressing, or diverting them during critical periods and subsequently engaging with them as necessary during calmer phases when their focused attention is not essential for success. Ultimately, however, a team that is functioning optimally will be able to respond appropriately to the ebb and flow of each team member's attentional focus. Currently, such sensitivity is typically acquired through years of experience working with one another, but we could foresee a time when such adjustments to individual differences in attentional state are augmented with additional information from behavioral and physiological assessments of individual mental states carried out in real time during training or during real surgical scenarios.

2) External Distractors: With respect to external distractions, these vary in terms of their relative predictability or constancy as well as their importance across individuals. As confirmed in interviews with clinicians, certain noises in the OR such as the ECG beeping or the sound of a patient's respirator are repetitive and either constant or sufficiently predictable that they merge easily into the auditory scene. This allows the OR team to effortlessly filter them out, much like ignoring background household noises. Similarly, certain visual properties, although they are described less often, may also fade into the background of the overall visual context. Such factors may still affect performance but not due to distraction. For example, bad lighting in the OR might be seen as visual distractions and can lead the team members to drift away from their tasks, but as long as it is constant it is not likely to serve as a distractor in the general sense of the term.

Other external events may still be at least somewhat predictable, including, for example, the possibility for unanticipated heavy bleeding or rapid drop in blood pressure during certain phases of specific surgeries. These events might suddenly require the immediate and full attention of the entire team, or their demands may land on one or a few individuals with specific responsibilities. However, because all of these events are at least somewhat predictable, they, too, may not need to be considered distractions but rather simply part of the ebb and flow of the demands of surgery. Indeed, it may be possible to include such events in training scenarios, allowing an individual or team to gain experience with coping with them, learning to work together better, and potentially reducing stress in a similar situation in the future.

Other external events are more transient and essentially entirely unpredictable. They do not belong in the scene and may cause the team's attention to the primary task to drop immediately. Again, the effects of these events may fall unequally on one team member, or may affect the entire team simultaneously. For example, as mentioned before, surgeons in an OR are often ward physicians, meaning that they are still responsible for their ward in case something unexpected happens there. Therefore, surgeons tend to be very sensitive to the sound of their beeper, which might signal a need for them to respond to events on their ward. This sound is described by our expert as the most annoying and distracting sound in the entire OR, because it means that the surgeon now has to let go of the current task and cognitively deal with the events on the ward. Although this is evidently distracting for the surgeon, even a brief lapse in the lead surgeon's attention can impose an increased cognitive load and strain on the entire team. Furthermore, unlike mind wandering (i.e., internal distractions), these calls or pager distractions can be equally possible during a boring phase of the operation or, at times, during critical phases. Similarly, our experts described occasional opening of the doors of the OR as particularly distracting, as may be movement of personnel around the room. However, these activities are also often be necessary and unavoidable as personnel enter/depart.

3) Team-Related Distractors: Team-related distractions in the OR can be induced by *unexpected* internal or external events as described previously, but may also arise from *expected* and even necessary demands on the distribution of attention. Surgeons often highlight the importance of effective teamwork during surgeries, emphasizing that unfamiliarity among team members can lead to suboptimal responses and coordination (e.g., miscommunication). Unexpected distractions are typically transient events that happen suddenly during the surgery, and although they cannot be avoided, there are strategies that could be taken to minimize their effects. Secondary tasks such as carrying on irrelevant conversations, or teaching students or residents, or simply not having a good understanding between team members, on the other hand, are effects that unfold over a longer period of time, and may engage multiple members of the team to different degrees. These kinds of team-oriented distractions cannot always be avoided and although they do not

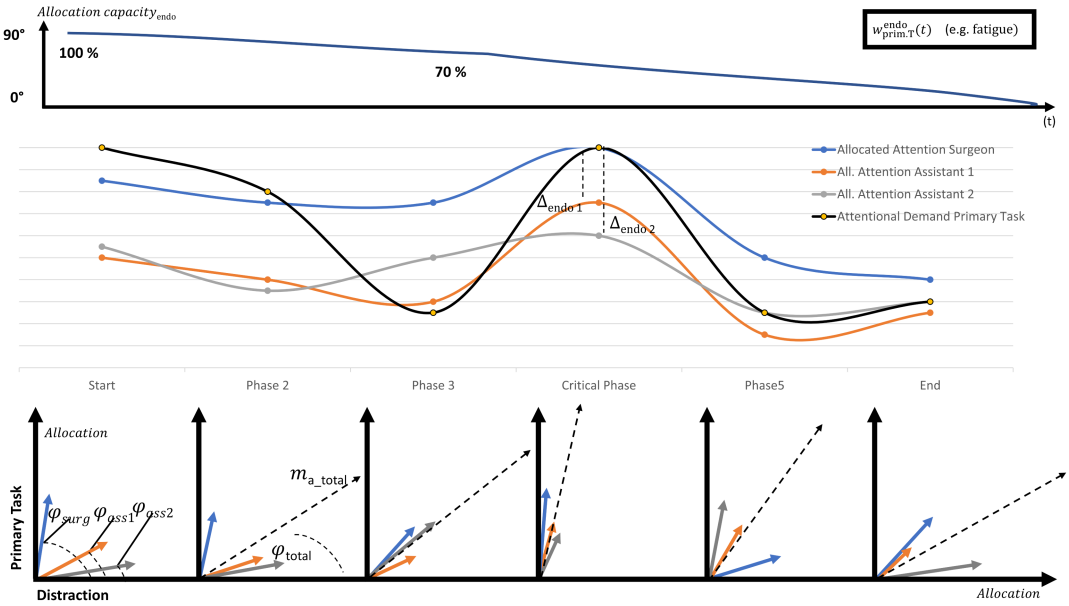


Fig. 2. Variation in attentional demand during episodes of surgery. A critical phase may be, for example, the engagement of critical anatomical structures or periods of bleeding that must be managed, or similar. The lower panel shows the corresponding attention allocation vectors (blue, red, and gray arrows) and the upper part of the figure visualizes the attentional demand of the team members during stages of surgery, which can be at different levels Δ_{endo} for, e.g., the surgeon and the assistants. Projections, later called $a_{\text{Prim.T}}$ and $a_{\text{Distr.}}$ of the vectors describe the part of attention provided towards the actual task and the distractions contribution to the attentional attentive distribution (see also Fig. 3). The instantaneous angle and magnitudes of the projections derived from the attention allocation vector are varying factors over the course of surgery and illustrate the idea of the models' competing forces between exogenous and endogenous attention allocation. Available attention capacity toward the primary task might be modeled by weighting factors $w^{\text{endo}}_{\text{primT}}(t)$, reflecting the team members' level of routine or fatigue over time $t \in \mathbb{R}$.

cause a sudden drop in the attention of the OR team they may still lower the overall attentiveness of individual members or the entire team. Therefore, it is also important that the model be able to represent the distribution of attention over a longer period of time.

4) Timing of Distraction: In addition to the source of the distraction, it is important to be able to model the timing of distractions within the course of a surgery. Surgeries usually consist of standardized procedures with periodic critical phases (e.g., approach or engagement of critical anatomical structures or periods of bleeding that must be managed) and boring phases in between. While some phases, both critical and boring, may be anticipated, both may also occur in a unplanned manner. Moreover, the ebb and flow of critical versus routine episodes during the course of a surgery may vary among team members. Periods of great significance to the anesthetist may be virtually irrelevant to a radiology technician on the same team. Current findings validate that as task performance progresses, the frequency of mind-wandering escalates, suggesting its involvement in within-task psychological shifts, such as heightened boredom and deteriorating behavioral performance as time on task extends [52]. Matching results showed fatigue and tiredness to be the most common distracting factor, contributing to errors around scrub nurses for instance [53]. Nevertheless, while individual team members' attention may be free to wander (i.e., be subject to distraction) during some phases of the operation, the attention of the entire team must be retrievable in fractions of a second during such a critical phase, see Fig. 2, for a diagram of these different phases.

5) Fatigue: As we consider how susceptibility to distraction may change over time, we must also consider how individual mental states may change. In particular, we must consider fatigue. From fields such as road traffic, aviation, and even military operations, a variety of studies have shown that fatigue due to sleep deprivation or sleep suppression impairs brain activity important for making moral decisions in dynamic and stressful situations (e.g., driving, reporting, or interacting with people), and insufficient sleep has been shown to have effects on performance and cognition [17], [18], [35]. Lack of sleep is an important factor in the medical field [17], [23], [35], [41]. Fatigue accumulates over time, and performance can still be impaired after the person has slept sufficiently [42]. However, fatigue does not merely build up over time, it is thought to have a rather complex connection with circadian rhythms and local brain mechanisms involved in cognitive fatigue [54]. Although there is an interaction, factors are distinguished that affect the body (physical fatigue) and the mind (mental fatigue) [55]. Mental fatigue, for instance, is considered one of the most significant causes of accidents in modern society [56]. OR-team members, like most individuals in high-stress jobs with irregular schedules or night shifts are especially vulnerable to depression and other mood disorders, as they face both mental- and also physical fatigue, which relates to physical work, incorrect posture, repetitive motion, or long static loading patterns [20], [21], [22], [40], [57].

6) Summary of Distractions in the OR: Developing a better understanding of the nature and impact of distractions on performance during surgery is a critical first step toward developing

training systems for enhancing individuals' resilience to these distractions, and, eventually, developing digital assistants to mitigate their effects during surgery.

By developing a model of how attention functions in surgery, it may be possible to develop training methods that more accurately reflect the actual demands imposed by these inevitable but unpredictable situations. In addition to training methods, technology can play a pivotal role in assisting surgical teams with managing distractions and maintaining focus. For instance, implementing electronic signaling systems or smart door mechanisms can provide visual or auditory cues to indicate when it is appropriate to enter or exit the OR. These signals can be synchronized with the progress of the procedure or other contextual factors, ensuring that interruptions are minimized and team members can enter the room safely without disrupting the surgical flow. By leveraging technology in this manner, surgical teams can better manage external distractions and optimize their attentional resources for patient care. To achieve these aims, a robust model facilitating the generation of hypotheses is indispensable. Such a model enables systematic studies in controlled laboratory settings or real-world operational environments, providing insights into the dynamics of distraction effects and strategies for coping with them. By investigating these aspects, we can develop comprehensive approaches to optimize performance amidst distractions, ultimately fostering productivity and well-being in surgical context.

III. Modeling Attention and Distraction—Surgical Context

A. Existing Attention Models for Clinical Environments

Perhaps the most studied model in the clinical environment is the model of visual scanning called salience, effort, expectancy and value (SEEV) [58], [59], [60]. The SEEV proposes that a linear weighted sum of these four factors can predict the degree of attraction of visual scanning to a particular area of interest (AOI). The net attractiveness of each AOI can be calculated with respect to all other AOIs including the one currently in view (i.e., the case that the gaze does not shift). Based on this overall weighting, the model can predict whether or not gaze will shift and, if so, to where it will shift. The SEEV was shown to be applicable in certain clinical use-cases [61], [62], [63], and models like the SEEV suggest ways in which incorporating psychophysiological markers related to arousal and attention could contribute to a more specific model of attention in ORs. However, while such models are already valuable, they do not necessarily fully encompass all the factors that should be taken into account within the surgical context. In particular, the SEEV is limited to visual information both in terms of input (the AOIs) and response (shifting of gaze). Visual information and gaze direction, while essential in the surgical context, encompass only a subset of the possible external distractors mentioned by surgeons and others, and there is as yet no mechanism within this model to accommodate internal distraction, nor is there a clear representation of the way that distraction unfolds over time. By incorporating these more general considerations, a model of attention and distraction during surgery should be able to not just account for specific distraction events, but also predict how future distractions may affect behaviors based on the current state, in both simulated and real environments. This goes in hand with a study from Stevens et al. [64], which showed that the second-by-second neurodynamic profiles [from electroencephalography (EEG) data dynamics] of OR teams were comparable in both the simulation and a live patient environments. Efforts to bridge this gap are crucial, emphasizing the importance of validating cognitive dynamics observed in simulations within live environments. To achieve a better understanding of attention and distraction, future studies should focus on integrating comprehensive models that consider contextual factors and validating findings across simulated and real-world settings, facilitating effective coping mechanisms in surgical scenarios. These models should further consider various factors, including data from psychophysiological markers, to accurately depict attention processes, particularly in surgical contexts.

B. Two-Competitor Model

The proposed model was previously used to address the challenge of defining and measuring immersion in the context of rapidly advancing immersive technologies [30]. It aims to provide a clear framework for understanding how attention operates within immersive experiences, thereby facilitating research on the human aspects of immersive interfaces. The model conceptualizes immersion as involving the projection of attention into an induced reality, building upon established taxonomies of attention. By elucidating the relationship between attention and immersion, the model lays the groundwork for assessing and optimizing immersive experiences using established psychophysiological measures. By incorporating a two-competitor model, which quantifies the interplay between different modes of attention, the approach may further provide a measurable way to analyze attentional processes during surgery. This model not only highlights the importance of managing distractions but also emphasizes the link between attention and surgical performance. Therefore, adapting this approach to the surgical context could enhance training methods, enabling surgical teams to better cope with distractions, maintain focus, and optimize their performance in the OR.

Adapting this model to attention and distractions in surgery would involve redefining the sensory streams and parameters to reflect the specific stimuli and factors present in the surgical environment (see Fig. 3) for a detailed graphical representation.

1) Influence of Distracting Events: In the OR setting the individual attention control of each OR team member may be seen as competing against powerful internal and external distractions that threaten to capture attention [30], [65], [66], [67]. These counterparts are represented in Fig. 3 by distraction on the x-axis and the actual (surgery related) task of the OR members on the y-axis. Both dimensions reflect the exogenous and endogenous attention type. However, we may assume an imbalance here, where the y-axis (primary task) is mostly driven by endogenous attention and the x-axis (distraction) by exogenous bottom-up attention (i.e., the "uninteresting" and irrelevant but

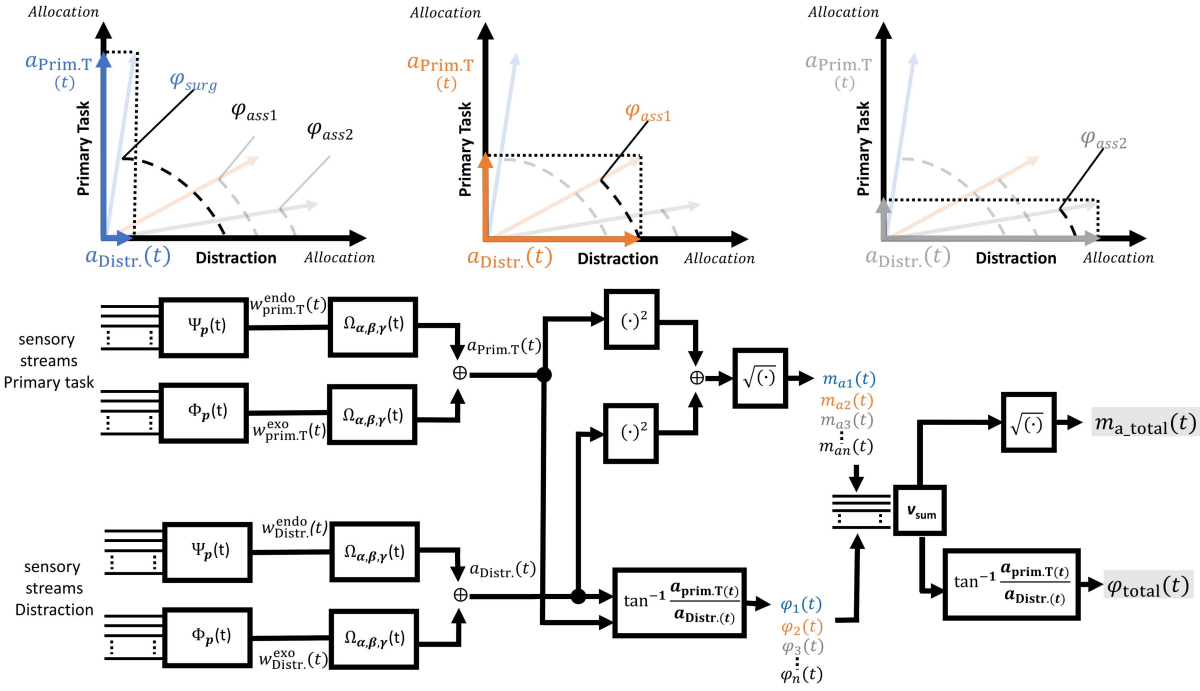


Fig. 3. In surgery, sensory streams would include First: visual and auditory inputs relevant to the procedure, such as the patient's vital signs, surgical instruments, and communication among team members and Second: distractions including various factors such as alarms, noise from equipment, interruptions from personnel, environmental changes, or unexpected events. Parameters p in the functions represent various factors influencing attention and distractions during surgery. These could include the complexity of the procedure, the urgency of the situation, the noise level in the OR, interruptions from personnel or equipment, fatigue, stress, and the surgeons level of expertise. The output angle $\varphi_i(t)$ indicates the degree to which the team members' attention is focused on the surgical task at any given time. Fluctuations in this angle can reflect shifts in attention caused by distractions. The magnitude of the attentional focus vector $m_{\mathrm{a},i}(t)$ represent the intensity or strength of the team members' attention directed toward the surgical task. Distractions may cause fluctuations in this magnitude, affecting the team members' ability to maintain focus. Angle $\varphi_{\mathrm{total}}(t)$ and magnitude $m_{\mathrm{a_total}}(t)$ of $\boldsymbol{v}_{\mathrm{sum}}(t) = \sum_{i=1}^{N} \boldsymbol{v}_{\mathrm{a},i}(t)$ aggregate the individual attention vectors into one collective team attentional state vector.

attention grabbing physical stimuli mentioned before). However, distractions with personal relevance that drive endogenous attention are possible in the distraction domain as well, such as interest-stimulating information about a patient of particular significance to the surgeon at the ward. The magnitude and angle of the attentional focus vector of a team member i at time t represent the current state of the time-dependent attentional allocation distribution between the stimuli related to primary task and the distracting stimuli and can be defined as

$$\boldsymbol{v}_{\mathrm{a},i}(t) = m_{\mathrm{a},i}(t) \begin{bmatrix} \cos\left(\varphi_i(t)\right) \\ \sin\left(\varphi_i(t)\right) \end{bmatrix}$$

where fluctuations in the magnitude due to endogenous and exogenous attention grabbing distractions modulate the resulting vector sum: $\boldsymbol{v}_{\mathrm{sum}}(t) = \sum_{i=1}^{N} \boldsymbol{v}_{\mathrm{a},i}(t)$, affecting the cumulative attentional strength toward the surgical task.

The endogenous force along the y-axis serves to maintain the attention allocated toward the goal task, whereas the attraction of attention by distracting stimuli pulls it away. When the attention demanded by distracting stimuli rises above the capacity of the endogenous attention directed toward the goal (y-axis), the team member's attention has been captured by the distraction and they can no longer screen it out. In this context the endogenous part of the model is influenced by different factor such as the internal motivation of the team-member and the effect of fatigue. In addition, the demand of attention for a certain phase of the surgery plays an important role for each individual team member as shown in Fig. 2.

2) Influence of Fatigue on Endogenous Attention: The term fatigue describes the persistent feeling of being tired, exhausted, or listless [68]. In surgery, fatigue is a multifaceted phenomenon and the strategies to mitigate its effect during surgery do not follow a standardized procedure nor are they universally adaptable to all clinical situations and OR-team members. The combination of fatigue and sleep stressors often complicates identifying what is primarily influencing subjective fatigue. Although, there are models which aim to describe effects of fatigue [69] in certain areas, up to now, there is no standardized way to objectively measure it. For this reason, the measurement and the assessment of fatigue needs to be tackled on a more individual level, supported by environmental and systemic changes [35], [41]. Integrating fatigue into the two-competitor model involves considering how fatigue influences attentional processes and affects the allocation of resources within the model. Fatigue can act as a modulating factor that influences the parameters of the two-competitor model. For instance, fatigue levels and sustained attention could affect the sensitivity of attentional processes or the rate of attentional resource depletion over time [70]. Furthermore, fatigue introduces competing demands for attentional resources. As fatigue increases, attentional resources may become more limited, leading to difficulties in maintaining focus

and resisting distractions [71], [72]. Also, fatigue can impair attentional control mechanisms, making it harder to regulate attentional focus and switch between tasks efficiently. This could result in decreased performance and increased susceptibility to distractions. The interaction between fatigue and attention may be dynamic, with fatigue levels influencing attentional processes, and attentional engagement or disengagement affecting fatigue levels in return. This bidirectional relationship could be modeled within the framework of the two-competitor model. The model could incorporate feedback mechanisms that account for the influence of fatigue on attentional processes and vice versa. For example, fatigue-induced lapses in attention could lead to increased fatigue levels, creating a feedback loop that exacerbates attentional difficulties.

3) Influence of Experience on Primary Task: Considering experience, experts are more likely to be able to block out distractions during surgery and are less at risk for large mental workload and stress [73]. For instance an experiment that tested how attention is divided between the surgery and outside distractions was performed in surgeons, and the performance was compared between expert and novice surgeons. The tasks were two standardized auditory distracting tasks (easy and difficult), requiring responses during surgery. Outcome measures were the number of correct auditory responses, task time, number of surgical errors, and instrument movements, as well as error rate, error probability and hand specific differences. The junior surgeons were less able to focus their attention and performed worse during these divided attention conditions. Unexpectedly, the correct responses to simultaneous auditory stimuli were less frequent in experts compared to novices, which showed a higher distractibility of younger colleagues from the important primary task [74]. The authors explain it with the experts deliberate neglect of the auditory task in favor of greater attention to the laparoscopic task. Interestingly, other studies have reported that through extensive practice, experts develop spare attentional capacity for simultaneous tasks and a higher endogenous attentional control [73].

This may go in hand with Fristons' proposed free energy principle, which says that the brain always make predictions to reduce surprise [75]. Sensory inputs are used to update and align an internal model of the world constantly. The free energy, bound to surprise is minimized by changing the predictions. In the context of surgery, predictions made by experienced surgeons might be much more accurate because they have encountered many different situations and, thus, have built up a repertoire of predictions of what might happen, allowing them to cope with upcoming events more quickly and more accurately then their less-experienced colleagues. That is, their internal representation of what is currently happening and what is likely to happen next may need only a few updates to find the correct solution while novices need far more iterations. Within the two-competitor model, this could be modeled in terms of a lower fluctuation (or lower entropy) of $\varphi_i(t)$. Thus, lower entropy in $\varphi_i(t)$ and increased weighting of endogenous attention via can serve as quantitative markers of predictive precision and task-focused experience. See [76] for further links between free energy minimization and the two-competitor model.

4) Incorporating Flow-State (FS) and Teamwork: According to [77], when a person engages in a task that presents an optimal balance between challenge and skill, they may enter a "FS", a state in which they are fully engaged in the task and yet the task feels almost effortless. In surgery, we might consider that one goal is to achieve an FS from the interaction between OR team members, i.e., the surgeon, assistant, scrub nurse, and surgical technician and the demands of the particular surgery, or at least for the surgeon to achieve that state [78]. The concept of "flow," i.e., the attentional focus vector "sticks" effortlessly to the y-axis in Fig. 3, is important to the present discussion of the two-competitor model in surgery for two reasons. First, the ability to enter an FS likely improves with experience. A proficient surgical team learns to work together and ideally over the course of multiple surgeries they come to understand each other implicitly. For example, in a well-coordinated team, the scrub nurse knows exactly when the surgeon needs a certain device to proceed. However, the demands of assembling surgical teams when different personnel have different work and vacation schedules often mean that teams are made up of less well-acquainted members and are put together with little predictable pattern. Flow distractions (FD), causing the attentional vector to tip over to the x-axis in Fig. 3, in the sense of traffic flow are often related to small layout-related disruptions on equipment-related disruptions for instance, and these minor FDs can sum up and lead to an increase of major FDs [79]. By developing a better understanding of how attention and FS functions within and across surgical team members during surgery by the tow-competitor model, we may be able to develop tools and strategies for improving the achievement of an FS even within surgical teams even when they are not so familiar with one another.

Second, we need to understand how attention operates in surgery because an FS depends on an optimal balance between the challenge of the task and the skill of the performers, and thus, can be disrupted both by demands on attention that exceed available capacity (i.e., distraction) and by the gradual loss of attention due to, e.g., fatigue or a lack of engagement (too little challenge) (see Fig. 2). Over the duration of a surgery, surgeons are likely to be more cognitively challenged than the assisting staff, which may help them stay engaged and "in flow." In contrast, assistants' attentional focus and/or anticipation for certain situations may decrease more quickly than the lead surgeon's because their role is less fully engaging to them, causing them to lose the FS while the lead surgeon may still remain in it. This disparity in task demand and its effect on attentional control needs to be taken into account for the later model, but for now we will focus on the problem of distraction, that is, the circumstances in which some event causes the surgeon or surgical team to redirect their attention away from their current task.

IV. Challenges and Opportunities

The development of robust models that capture the nuances of attention allocation during surgical procedures presents challenges but also interesting future opportunities. Coupled

with advanced measurement technologies capable of accurately monitoring attentional states in real time, these models serve as the foundation upon which smart assisted surgery can thrive. As we explore in more detail the question how attention operates within the surgical context, we unlock new avenues for innovation and collaboration between humans and machines. We foresee a future use of these models of attention and distraction in surgery to improve medical practices by enhancing surgical precision and patient outcomes. These models could potentially integrate advanced technologies like artificial intelligence (AI)-driven attention monitoring systems, personalized distraction management techniques, and immersive virtual environments tailored to individual patient needs. This holds the potential to optimize surgeon focus during procedures, minimize distractions, and mitigate errors. However, achieving this vision entails overcoming various challenges, including regulatory hurdles, ethical considerations regarding patient privacy, and the need for extensive validation to ensure safety and efficacy. Another factor is the need for suitable objective data which can be incorporated into the models and serve as the backbone for attention allocation recognition. Nonetheless, amidst these challenges lie significant opportunities for innovation and collaboration within the medical community to refine these models, leading to improved surgical experiences and better patient care.

A. Challenges

1) Need for Suitable Experimental Data: Discovering objective biomarkers that align with attentional effects is paramount for enhancing the accuracy and efficacy of these models. While current measurement outcomes, in our view, do not furnish all the requisite data, they do reveal markers that could be assimilated into initial models. Various measurement techniques have been developed to aid in achieving important objectives in the surgical context and are mostly related to stress responses [1], [2], [3], [6], [7], [16], [31], [32], [33], [34]. Some studies collected data from EEG, pupil dilation, heart rate, and breathing rate, as well as force exerted by the subjects' thumbs while operating instruments [80]. Such physiological and behavioral data may be correlated with, for example, skill level, fatigue, or even distraction. However, we are still lacking an overarching model of how these physiological and behavioral measures, and the affective and cognitive states that they supposedly index, relate to these more complex theoretical constructs.

2) Need for More Advanced Assessments of Attention: Existing research related to attention during surgery mostly assessed the distribution of visual attention using eye tracker devices, i.e., visual attention task pre- and postsurgery or before and after extended shifts (12–24 h) [81], [82]. A more recent study showed that more fixations of eye gaze to areas of interest on the patient monitor appeared during critical than noncritical scenarios, while dwell time did not differ and the professional experience did not significantly influence the visual attention [83]. Besides physical measurements during surgery, attention tests, and questionnaires can give further insight into a surgeon's attention levels. For instance, the efficiency of a novel mental skills curriculum was shown to improve the attention management of novices significantly if it was used in parallel to laparoscopic training [73].

3) Need for Nonintrusive Measurement Approaches: In addition, due to the measurement setting, namely the OR, there are limitations in terms of the applicability of measurement technology. The fact that the OR team works in a sterile environment and should not be influenced by the sensors makes measurement in action challenging. Measurements approaches need to change from intrusive to minimal and nonintrusive methods such as video based data collections or smart textiles [44], [84], [85], [86], [87]. Further minimizing the size of the measuring devices will help gaining the acceptance and lead to a more frequent use in such environments [88], [89]. Obviously, these approaches have the need to get validated against gold-standard devices, but have already shown promising results [90].

4) Incorporation of Existing Correlates From Stress and Fatigue: Recent studies have highlighted various physiological markers for assessing stress and performance levels during surgical procedures. For instance, heart-rate-variability (HRV) has emerged as a valuable tool for objectively evaluating stress induced in workplace settings, enabling the identification of stressors during operations and discerning differences in stress levels between surgeons in different roles [38]. However, while HRV can capture short-term effects, further research is needed to uncover patterns related to long-term stress effects, prevention strategies, and recovery. Intraoperative EEG and HRV measurements have been utilized to gauge stress factors during surgery, with parameters such as heart rate and EEG waves correlating with stressful situations and distinguishing between experienced and less experienced surgeons [91]. In addition, studies have linked ECG measurements to acute mental stress and surgical performance, demonstrating higher stress levels in relation to specific physiological responses [92]. In laparoscopic surgery, a combination of EEG, eye-blink times, fNIRs, and cardiac data has been employed to monitor cognitive load, revealing a decrease in load with surgeon proficiency [50]. Notably, subjective task difficulty and completion time have shown correlations with physiological features, with multimodal approaches offering superior stress predictions compared to individual markers alone. Studies have also explored the potential of neuromonitoring findings to classify surgical expertise levels, demonstrating distinct activation patterns in the prefrontal cortex among novices and experts across various tasks [80], [91], [93]. In addition, a novel EEG biomarker, VC9, has been proposed for assessing cognitive load and performance, showing promise in discriminating between different levels of cognitive demand in surgical simulations [94]. Nevertheless, there are recent efforts in monitoring long-term factors such as impacts of the circadian rhythm on an individual's moods. Unfortunately, these factors are hard to evaluate since they heavily depend on individual social dynamics, schedules, and temperament [57].

Some of those results might be integrated in the development of the two-competitor model. However, little focus was directed to actual measurements of the overall effects that could represent distraction and attention and, thus, serve as a basis for a neurodynamic model or a supportive monitoring system. A lack of suitable attention research is evident, but

cross-disciplinary insights from fields like neuroergonomics, cognitive neuroscience, human–computer interaction, and other application areas such as automotive and aviation might inspire future experiments and modifications to the model, contributing to a deeper understanding of attention effects in realistic surgical settings or even throughout real clinical scenarios [42], [60], [95], [96], [97], [98], [99].

5) Smart Assistance Technology Needs to be Accepted: The successful adoption and integration of smart assisted surgery hinges not only on technological advancements but also on the acceptance and cooperation of the surgical team. Surgeons, nurses, anesthesiologists, and other members of the surgical staff play pivotal roles in embracing these innovations and leveraging them effectively to enhance patient care. Despite the potential benefits of smart assisted surgery, there exist significant barriers to its widespread adoption within the surgical community. Chief among these challenges is the need to cultivate trust and acceptance among the members of the surgical team toward these emerging technologies. The challenge of societal acceptance of new technologies is a major issue in the field of collaborative technology development and the various strands of research and practices summed up under the term of Ethical, Legal, and Social Issues (ELSI) research. ELSI is commonly employed as an encapsulation of a diverse spectrum of research endeavors that explore the multifaceted effects of science and technology on society [100]. The notion of acceptance of new technologies has a problematic connotation as it is partly equaled with the attempt to arrive at high individual acceptance levels of new technologies. This perspective is being criticized for narrowing down the issues of acceptance of new technologies to observing and tackling technosceptic attitudes on an individual scale [101]. The primary critique of this limited view of acceptance revolves around the suggestion that the relationship between society and technology is singular, based on the myriad of individual user-technology interactions. One perspective on this issue views the society with all the elements that are ascribed to it including values, norms, or institutions, while a contrasting point of view considers technology as having attributes attached to it, such as clear causalities, controllability, and effectiveness. A better approach may be the employment of the perspective of socio-technical interactions, which transcends the dichotomy of technology versus society [102], [103]. Instead of asking how to bring technology into society with both of them considered as independent, separate realms, the perspective of socio-technical interactions analyses technology development as a process of interplay between different professional (techno-)cultures, which shapes the aims and appropriate uses of a new technology within the context of the society, which will use it. Similarly, in a socio-technical interaction the designated users of technologies develop ways to interact with the technology which are, by definition, not fully determined by the technology's usability or design. Thus, the acceptance and subsequent diffusion of a new technology, which play a role in its "success" or "failure," should be studied in situ within the socio-technical interactions of their development (e.g., laboratories, research units of organizations, experimental spaces) [104]. The socio-technical interactions of technology development and deployment are often marked by heterogeneous knowledge and normative structures, which can lead to contradictory dynamics and development, such as the "paradox of perfection" or the Collingridge Dilemma [101]. When acknowledging the reciprocal effects between technology and society, in other words, the resulting coevolution between newly introduced technologies and the organizations that utilize them, it is important to include relevant stakeholders in the introductory process from the very beginning [105]. This has also been discussed in the field of human–computer interaction under the term "value-sensitive-design" [106]. To consider relevant questions of discriminatory effects and questions of ethics and data protection it is beneficial to include stakeholders with different cultural backgrounds, abilities, and organizational roles in the process to gain a diverse view and take into account also hidden implications of specific needs through a diversity panel [107]. New technologies in the surgical field shall match the attitudes, preferences, and routines of individuals, but these technologies might not be accepted if they do not fulfill some requisite characteristics. First, they should form a vital part of the social practices of surgical teams (team-level). Second, the explicit and implicit rules of organizations such as surgical wards or hospitals should be fulfilled. Finally, their implementation should be in accordance with legal norms on the societal level such as the general data protection regulation as well as with societal values such as accountability and responsibility.

B. Opportunities

1) Improved Acceptance of Smart Assistance Technology in Surgery: In the rapidly evolving landscape of modern medicine, the integration of technology within surgical practices has become increasingly prevalent. Among these advancements, the concept of smart-assisted surgery stands out as a transformation approach that holds potential for enhancing surgical outcomes. At its core, smart-assisted surgery harnesses the power of AI and advanced monitoring technologies to provide real-time assistance and decision support during surgical procedures. By leveraging AI algorithms and sophisticated sensors, these systems can analyze vast amounts of data, including vital signs, anatomical images, and surgical parameters, to offer invaluable insights and recommendations to the surgical team. Benefits evolving from these technologies are sometimes hard to explain to the user and even harder to compare between studies. Leveraging the acceptance of these technologies has the need for a tool, which is able to give an objective overview about benefits or drawbacks of new approaches. The two-competitor model might be used as such a tool for cross-study comparisons.

2) Comparison and Evaluation of New Technologies: The model could offer an objective evaluation by leveraging data-driven insights to assess the strengths and weaknesses of study designs, aiding researchers in selecting the most suitable approach for their research objectives. With its predictive capability, it could enable researchers to anticipate potential outcomes, facilitating the design of studies that maximize the likelihood of

meaningful results. In addition, its adaptability ensures incorporation of updated information and refinement of recommendations in the dynamic field of surgical innovation. By guiding researchers toward efficient study designs, the model might optimize resource allocation, ensuring judicious use of limited resources to yield valuable insights into the safety, efficacy, and usability of new surgical technologies.

V. Conclusion

The two-competitor model of attention offers valuable insights into cognitive processes within dynamic environments like surgical settings. It provides a framework for understanding the multifaceted factors influencing attention, including external distractions and contextual variables specific to surgery. In addition, the model facilitates the development of tailored interventions and coping strategies to optimize performance and mitigate distractions in practical scenarios. To improve the model, future efforts should focus on integrating suitable advanced measurement systems for psychophysiological markers into simulated surgical interventions that are subject to controlled distractions and human–machine interactions. This data can then be transferred to an AI model for training, whereby suitable attention markers are obtained to enhance the two-competitor models' specificity and accuracy in depicting attentional processes. Seamless integration of simulated and real-world data is also crucial for ensuring the model's validity and applicability across different environments. This integration will advance our understanding of attention and distraction dynamics, ultimately leading to more effective interventions in surgical settings. Moreover, recent advances in neurotechnology present opportunities for real-time monitoring and interventions to identify factors affecting the performance of the OR team. However, some measurement methods, such as neural attention effects and cortico-muscular interaction, require further investigation in the surgical context. Moreover, drawing upon cross-disciplinary findings from domains such as space exploration, aviation, and human–machine interfaces can spark enhancements to the model and enrich our comprehension of attentional dynamics in surgical contexts, while on the other hand help the model to gain a more generalize concept of attention for highly digitalized domains. By addressing these aspects, the two-competitor model can serve as a valuable tool for enhancing performance and patient outcomes in surgical settings while advancing our broader understanding of attention and distraction dynamics.

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